Identify common factors

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: Find the highest common factor of 84 and 126. Copy or complete the first step.

2. List the common factors of 36 and 60.

3. Find the HCF of 96 and 144.

4. Miro says: Miro chooses the largest factor from either list rather than the largest shared factor. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the highest common factor of 84 and 126. Copy or complete the first step.
   Answer 42.

2. List the common factors of 36 and 60.
   Answer 1, 2, 3, 4, 6 and 12.

3. Find the HCF of 96 and 144.
   Answer 48.

4. Miro says: Miro chooses the largest factor from either list rather than the largest shared factor. Is this correct? Explain.
   Answer A common factor must divide both numbers exactly.

Identify common factors

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. List the common factors of 36 and 60.

2. Find the HCF of 96 and 144.

3. Two numbers have HCF 18. Give a possible pair greater than 50.

4. Identify and correct this misconception: Miro chooses the largest factor from either list rather than the largest shared factor.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. List the common factors of 36 and 60.
   Answer 1, 2, 3, 4, 6 and 12.

2. Find the HCF of 96 and 144.
   Answer 48.

3. Two numbers have HCF 18. Give a possible pair greater than 50.
   Answer 54 and 72 is one example.

4. Identify and correct this misconception: Miro chooses the largest factor from either list rather than the largest shared factor.
   Answer A common factor must divide both numbers exactly.

5. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Identify common factors

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Two numbers have HCF 18. Give a possible pair greater than 50.

2. Explain why this reasoning fails: Miro chooses the largest factor from either list rather than the largest shared factor.

3. Create a new example that tests the same idea as: Find the HCF of 96 and 144.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Two numbers have HCF 18. Give a possible pair greater than 50.
   Answer 54 and 72 is one example.

2. Explain why this reasoning fails: Miro chooses the largest factor from either list rather than the largest shared factor.
   Answer A common factor must divide both numbers exactly.

3. Create a new example that tests the same idea as: Find the HCF of 96 and 144.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.